

WHAT BREAKS FIRST AT 10x?

THE SPARK ANSWER THAT WINS THE \$200K OFFER

It sounds like a capacity question. It is a ranking question.

THE ANSWER THAT ENDS IT

(Capacity thinking)

- ✗ We would scale the cluster
- ✗ More nodes, more executors
- ✗ Autoscaling up

Nothing in it is wrong. It just ranks nothing.

THE ANSWER THAT GETS THE OFFER

(Rank by how cost scales)

THE COMPONENT

The shuffle, not the read.

WHY IT BREAKS FIRST

Shuffle bytes grow 10x. Partitions do not.

THE SIGNAL

Stage detail: spill appears, Duration Max above Median.

The shuffle grows either way. A hot key may not.

RANK BY HOW COST SCALES, NOT BY WHAT IS BUSY TODAY

10x data does not load every component 10x.

LINEAR AND PARALLEL

- Scans, filters, projections
- Money fixes these

WORSE THAN LINEAR

- Anything that moves data
- The shuffle lives here

NOT PARALLEL AT ALL

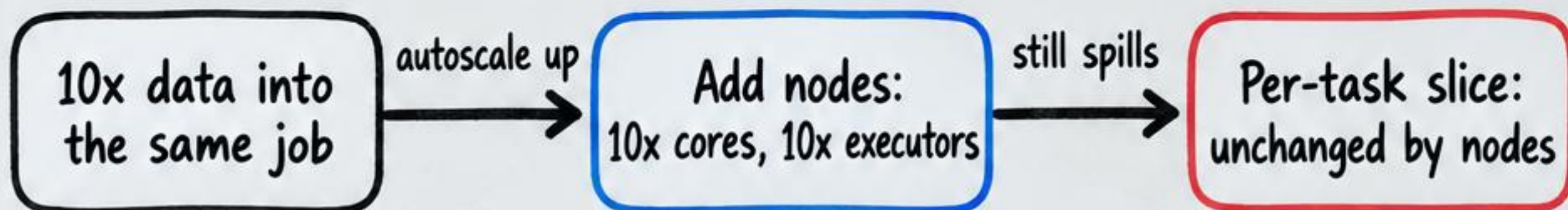
- One hot key, one task
- Worse - only if you have one

➔ ONE BOARD PER SENIOR MOVE

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MORE NODES AT 10x BUY PARALLELISM, NOT PER-TASK MEMORY

'Why not just add nodes?'



Each task's slice is set by the partition count, not the node count.

NODES BUY	NODES DO NOT BUY
More tasks running at once	More memory per task
A shorter queue for the stage	A smaller slice to hold
Throughput on core-bound work	A shorter skewed task

✗ **JUNIOR:** 'More executors, more parallelism, done.'

✓ **SENIOR:** 'Nodes buy parallelism, not per-task memory.'

More cluster answers a question nobody asked.

next: the one tier where more nodes still wins →

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WHEN SCALING THE CLUSTER IS THE RIGHT 10x ANSWER

'So is adding nodes ever the right answer at 10x?'

TIER	WHAT GROWS AT 10x	WHAT STAYS FIXED	THE MOVE
Linear and parallel	Bytes to read, 10x	Work per core	Buy cores. Money fixes it
Worse than linear	Shuffle bytes, 10x	Partition count	Move less data
Not parallel at all	Rows on the hot key	One task	Split the key, or redesign

HOW TO TELL WHICH TIER YOU ARE IN

- Cores busy, no spill: buy cores
- Spill at the median task: move less data
- One task long, the rest idle: hot key

More nodes is the right answer – for exactly one tier.

next: the worse tier that is still the wrong answer →

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AT 10x, RANK BY WHAT GROWS NO MATTER WHAT

'Skew is worse than shuffle. Why name the shuffle at 10x?'

THE TRAP: NAME THE WORST TIER

- ✗ A hot key can pin one task
- ✗ So you name the hot key
- ✗ But only if you have one

Guessing at a key you have not seen

THE RULE: NAME WHAT GROWS EITHER WAY

- ✓ Shuffle bytes grow with the data
- ✓ No hot key required
- ✓ True of any pipeline that shuffles

Rank by unconditional growth, not worst case

COMPONENT	GROWS AT 10x	CONDITION
Shuffle volume	Always	None
Spill	Once the slice outgrows memory	Partition count unchanged
Hot key	Only where one exists	You have a hot key

Worse is not first. Unconditional is first.

next: the read that turns a named component into proof →

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NAME THE COMPONENT AT 10x, THEN PROVE IT

'You named the shuffle. How would you prove it at 10x?'

WHAT 10x ACTUALLY CHANGES

500GB a day becomes 5TB a day

Shuffle bytes: 10x

Partition count: unchanged

Per-task slice: 10x

Task memory: unchanged

Once the slice outgrows the task, it spills to disk.

WHERE TO LOOK	WHAT YOU SEE	WHAT IT PROVES
Stage detail page	Spill stats appear	Slices outgrew task memory
Stage detail, Summary Metrics	Duration Max above Median	One task, not every task
Executors tab, Task Time	Totals per executor	Wrong read for a stage

Spill stats are absent, not zero, until a stage spills.

next: the fix that no cluster size can buy →

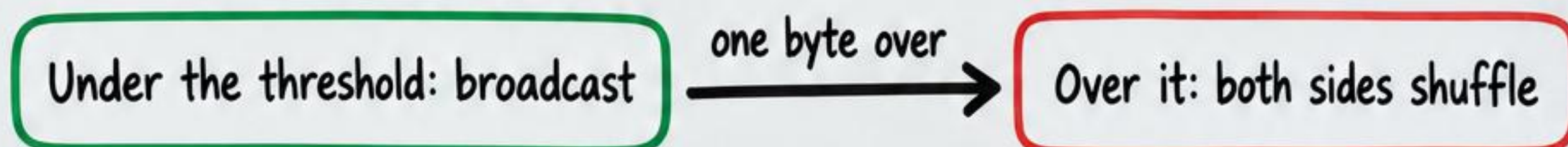
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THE 10x FIX IS A DATA-MOVEMENT CHANGE, NOT MORE NODES

'So what do you change, if not the cluster?'

COMPONENT	WHAT CHANGES AT 10x	THE MOVE
Shuffle	Bytes grow, partitions do not	More shuffle partitions, filter first
Broadcast join	One side may cross the threshold	Design for the shuffle, not broadcast
MERGE into Delta	More keys touch more files	Cluster on the merge key

BROADCAST IS A CLIFF, NOT A SLOPE



The planner picks sort-merge and shuffles both sides.
AQE can switch back to broadcast, but only after both sides have shuffled.

WHAT JUNIORS SAY HERE

- ✗ 'Small table, it will still broadcast.'
- ✗ 'It is a MERGE into Delta, that scales.'

Every fix here changes how data moves. None is a bigger cluster.

last: why the senior stops talking here →

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THE 10x ANSWER, IN ONE BREATH

'Why stop after one component, one reason, one signal?'

Rank by how cost scales, not by how busy it is today.
The fix is a data-movement change, not a bigger cluster.

- ✓ 10x does not load every component 10x
- ✓ Rank by what grows no matter what
- ✓ Name the shuffle. Prove it in the stage detail
- ✓ Move less data before you buy more nodes
- ✓ Unless the stage is linear - then buy cores

One component. One reason. One signal. Then stop.